SENTINEL

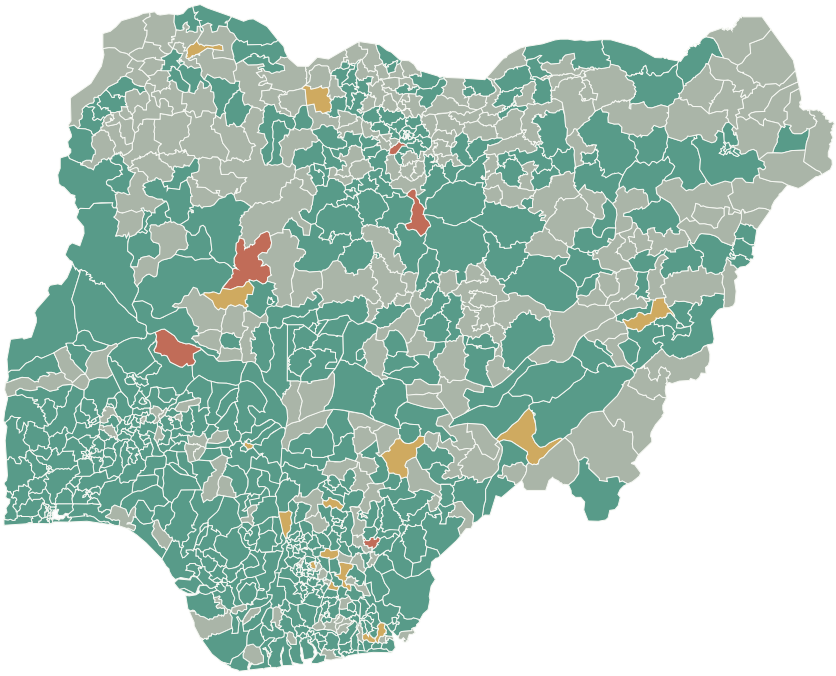
Mapping Nigeria’s digital divide through connectivity evidence and public-service locations

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Why we chose this problem

Our starting point was election result transmission: where should a team check connectivity before it needs to upload a time-sensitive document? That question led us to upload speed and latency, and then to a wider concern. The same need to send information matters to clinics and schools. We added these locations to the Ookla data to give the map a practical purpose: help a team decide where to investigate first. Because the observations are uneven, we built SENTINEL around verification visits and follow-up reports. We want to help users make a defensible next decision, without claiming that a historical speed test proves whether a site will work today.

What the available evidence shows



Insufficient evidence (234)
Poor observed performance (5)
Review performance (15)
Lower observed concern (520)

774 LGAs. Grey means missing evidence, not no service. Categories describe historical LGA averages, not current coverage, ward performance or election outcomes.

How we built the map

We matched the Ookla coordinates to LGA boundaries: 9,209 of 9,234 records matched; we excluded the remaining 25. We then calculated tile means and test totals, assigned each LGA to the state with the largest polygon overlap, and joined the health and education records by location. All 774 state assignments passed our 95% overlap check. We used upload speed and latency to rank measured areas, while keeping missing evidence separate.

Interactive map: inec-sentinel-omega.vercel.app
Code: github.com/De-Young07/INEC-SENTINEL

What we added to the supplied data

geoBoundaries: 774 LGA polygons plus state/FCT boundaries provide administrative context. Assignments use geometry and stable IDs rather than names alone.
GRID3 health facilities v3.0: 41,778 records; 6,004 missing/empty geometries excluded. 35,772 of 35,774 usable records match LGAs.
HOT OpenStreetMap education: 6,327 of 6,329 usable records match LGAs. These are education locations, not verified polling units.

Sources: geoBoundaries; GRID3 via HDX; HOT OSM education extract. Supplied Ookla file: Nigeria Ookla.gpkg. Source counts and matching rules are retained in the audit.

What we found and what we would do next

We cannot assess 234 LGAs from this extract. That is 30.2% of the 774 LGAs. We would start by collecting measurements in these areas. Calling them blackouts would go beyond what the data tells us.

The gaps matter beyond elections. Those 234 LGAs contain 10,308 matched GRID3 health records and 411 education records. We would work with local service managers to verify the sites and choose places for repeated tests. These counts help us identify where to ask questions; they do not tell us how many facilities are offline.

Our highest-concern results need checking on the ground. Madobi (Kano), Ezza South (Ebonyi), Doguwa (Kano), Rafi (Niger) and Edu (Kwara) fall in our poor-performance category, but each has just one observed tile. In total, 355 of 540 measured LGAs have fewer than ten. Edu also shows why we looked at uploads: its tile records 15.129 Mbps download but only 0.084 Mbps upload. We would repeat tests at different locations, times and on different carriers before recommending infrastructure.

How we turn the findings into a plan

Score = 100 x [0.70 x clamp((2.5 - upload Mbps)/2.5) + 0.30 x clamp((latency ms - 50)/200)], with each clamp restricted to 0-1. Scores below 35 indicate lower concern, 35 to below 65 need review, and 65 or more indicate poor observations. Thresholds are pilot assumptions, not validated BVAS requirements. Unknown areas receive visit priority 75, not a risk score.

We built the app so a team can search an LGA or state, inspect its measurements and matching details, and choose verification visits within a budget. We also added shared monitoring so a reported problem can have an owner and a record of recovery. The SMS/email adapters are prepared, but sending is not enabled.

What we still need to establish

We still need to confirm the supplied extract’s period and whether it represents mobile or fixed connections. Our averages give every tile equal weight, and facility records may overlap or be outdated. We have not validated population or socioeconomic layers, so we cannot yet compare need per resident or estimate beneficiaries. Schools are not verified polling units, and this map does not establish current nationwide coverage or the causes of poor performance. Our next step is a small field-validation pilot before making investment recommendations.